Research Square

Preprints are preliminary reports that have not undergone peer review.
They should not be considered conclusive, used to inform clinical practice,
or referenced by the media as validated information.

From News to Trends: A Financial Time Series Forecasting Framework with LLM-Driven News Sentiment Analysis and Selective State Spaces

Renjie Wang
Jilin University
Minghui Sun
smh@jlu.edu.cn
Jilin University
Limin Wang
Jilin University

Research Article

Keywords: Sentiment analysis, Time series forecasting, Mamba, LLM

Posted Date: April 17th, 2025

DOI: https://doi.org/10.21203/rs.3.rs-6277319/v1

License: This work is licensed under a Creative Commons Attribution 4.0 International License.
Read Full License

Additional Declarations: No competing interests reported.

Version of Record: A version of this preprint was published at Journal of Intelligent Information Systems on August 2nd, 2025. See the published version at https://doi.org/10.1007/s10844-025-00971-3.

From News to Trends: A Financial Time Series Forecasting Framework with LLM-Driven News Sentiment Analysis and Selective State Spaces

Renjie Wang[1,3], Minghui Sun[2,4*], Limin Wang[2,3]

[1]College of Software, Jilin University, Changchun, 130015, Jilin, China.
[2]College of Computer Science and Technology, Jilin University, Changchun, 130015, Jilin, China.
[3]Key Laboratory of Symbolic Computation and Knowledge Engineering of Ministry of Education, Jilin University, Changchun, 130015, Jilin, China.
[4]Research Institute of Jilin University in Shenzhen, Jilin University, Shenzhen, 518000, Guangdong, China.

*Corresponding author(s). E-mail(s): smh@jlu.edu.cn;
Contributing authors: wangrj23@emails.jlu.edu.cn;

Abstract

Stock price prediction is inherently challenging due to market volatility and the influence of external factors. Traditional forecasting methods primarily rely on historical price data, limiting their ability to capture market sentiment embedded in financial news. To address this limitation, we propose Senti-MambaMoE, a novel model that integrates historical stock prices with sentiment information extracted from financial news. Specifically, we fine-tune a DeepSeek-based large language model (LLM) for financial sentiment classification and incorporate the extracted sentiment information into our predictive framework. At the core of our approach is MambaMoE, which leverages the efficiency of state space models (SSMs) to model long-range dependencies while maintaining linear computational complexity, making it well-suited for financial time series forecasting. Additionally, the MoE mechanism improves the model's ability to capture diverse market behaviors by dynamically selecting specialized experts based on stock data patterns. Experimental results demonstrate that Senti-MambaMoE outperforms LSTM-based models by 23.7% and Transformer-based models by 6.3%, highlighting its superior performance in short-term stock prediction.

Keywords: Sentiment analysis, Time series forecasting , Mamba, LLM

1 Introduction

This study aims to evaluate the effectiveness of DeepSeek-based LLMs in sentiment analysis of financial texts and explore the potential of state-space models for stock price prediction. Time series forecasting models have become essential tools in financial decision-making, particularly in stock market analysis, which is inherently complex due to its high noise, time-dependent nature and sensitivity to external factors. Traditionally, stock price forecasting relies on technical indicators from historical data [1–3], which are limited to past information. In contrast, news titles provide insights into unforeseen events, policy changes, technological advancements, and public sentiment, enhancing the model's ability to capture factors beyond numerical data. By incorporating both quantitative and qualitative data, prediction models can improve accuracy and better reflect real-world financial dynamics.

Extensive research has been conducted in the field of sentiment analysis, particularly in the context of financial markets. Early studies employed models like LSTM [4, 5] and TextCNN [6] to analyze sentiment in text, focusing on the influence of investor sentiment on stock price movements. The introduction of the BERT [7] model marked a significant turning point, with pre-trained BERT-based models [8, 9] gaining widespread adoption for text analysis. In recent years, the rapid advancement of generative large language models [10] has further revolutionized a variety of tasks, demonstrating exceptional performance across multiple domains, including sentiment analysis. Recently, the High-Flyer Quant team open-sourced DeepSeek-R1 [11], along with several distilled, lightweight models based on DeepSeek-R1, such as DeepSeek-distill-Qwen-1.5b. DeepSeek leverages a Mixture of Experts (MoE) [12] architecture and Multi-Level Attention (MLA) to significantly reduce computational costs. Furthermore, having been trained on a vast corpus of Chinese text, DeepSeek demonstrates a profound understanding of semantic relationships in Chinese language processing, outperforming many models of comparable size in reasoning tasks. Notably, no research has yet explored the potential of DeepSeek distilled models for sentiment analysis in the financial text domain. Motivated by the advantages of DeepSeek, we aim to leverage these distilled lightweight models for sentiment analysis tasks in the financial context.

Financial time series forecasting serves as a critical foundation for decision-making across economic, infrastructure, and social domains. The primary goal of analyzing time series data is to decode the evolving relationships within complex dynamic systems. Current mainstream forecasting methods predominantly rely on autoregressive patterns extracted from historical data, demonstrating strong asymptotic consistency in stationary time domains. In stock price prediction, time series models remain the most widely used approach, including Autoregressive Integrated Moving Average (ARIMA), tree-based models, support vector machines, recurrent neural networks, and attention-based models. These methods leverage historical price data to predict

future price movements. With their ability to capture temporal dependencies, Long Short-Term Memory (LSTM)[4] networks and Gated Recurrent Units (GRU)[13] have gained increasing attention from researchers and industry practitioners. These models can learn complex nonlinear relationships within the data, enhancing prediction accuracy. Meanwhile, attention mechanisms[14] emulate human focus, enabling models to prioritize relevant information. Transformer[14], the most successful sequence learning model based on self-attention, has demonstrated superior performance in modeling long sequences, as evidenced by natural language processing experiments. The Bidirectional Encoder Representations from Transformers (BERT)[7], a pre-trained Transformer variant, has further improved performance over the standard Transformer model. In recent years, more advanced time series models have emerged. These include Transformer-based architectures like Crossformer[15], FEDformer[16], and PatchTST[17], as well as MLP-based approaches such as DLinear[18], TimeMixer[19], and TSMixer[20]. These models continue to push the boundaries of time series forecasting by leveraging innovative deep learning architectures.

State Space Models (SSMs)[21, 22] offer a strong balance of performance and computational efficiency by leveraging convolutional operations for capturing sequential dependencies while eliminating hidden states, allowing for parallel computation and near-linear complexity. The Mamba model [23] builds upon SSMs by integrating the Structured State Space Sequence Model (S4) [21] with a selective scanning mechanism (S6), achieving state-of-the-art performance in sequence modeling. This allows Mamba to capture valuable information similar to attention mechanisms, enhancing its application across domains like text and image processing. Mamba's combination of performance and precision, alongside its ability to focus on critical sequence features, makes it a promising candidate for financial time series forecasting.

In this study, we propose a two-stage model that integrates large language model (LLM)-based sentiment analysis with time series forecasting. By leveraging the news headline data from the Eastmoney stock forum, we embed sentiment information into time series data and employ a Mamba model with a mixture-of-experts (MoE) architecture to predict stock trends in China's stock market. Our contributions can be summarized as follows:

- We employ quantized LoRA (Low-Rank Adaptation) to fine-tune multiple large language models, including DeepSeek, Qwen, Llama and Bert, aiming to compare their performance in Chinese financial text classification.
- We propose an innovative time-series state space selection model that seamlessly integrates sentiment analysis for short-term financial time series forecasting. This model effectively captures both long- and short-term dependencies, extracting complex temporal patterns from financial data to improve forecasting accuracy and stability.
- Our model adopts a novel and efficient Mamba-based mixture-of-experts (MoE) architecture, where intra-patch experts handle local features while inter-patch experts coordinate global information. This hierarchical design enables the model to capture and process multi-level, multi-scale patterns in financial data, further enhancing prediction accuracy and robustness.

- Our model achieves state-of-the-art (SOTA) performance on three Chinese stock datasets, outperforming Transformer-based models by 6.1%, BiLSTM by 23.6%, and the original Mamba architecture by 43.4%.

2 Related works

2.1 Models for sentiment analysis

Yoon Kim[6] find that convolutional neural networks trained with pre-trained word vectors perform effectively for sentence-level classification tasks, including sentiment analysis. Regional CNN-LSTM[24], a model for dimensional sentiment analysis, predicting VA ratings by extracting region-specific affective information and integrating it across sentences. ConFEDE[25], a unified framework leveraging contrastive learning and feature decomposition to enhance multimodal sentiment representation. Lu et al. [26] introduce a novel electricity price prediction model that integrates bidding behaviors and market sentiment, fine-tuning a pre-trained LLM for improved spike prediction accuracy and real-time market forecasting. Wang et al.[27] propose a novel approach that combines Large Language Models and Generative Agents to enhance time series forecasting by integrating news events with time series data. Chen and Kawashima [28] study on large language model-based news sentiment analysis for stock price forecasting, comparing multiple models and demonstrating that adding sentiment data improves predictions. There are some initial attempts on evaluating LLMs for SA tasks. In addition, Wang et al. [29] conduct a preliminary study with ChatGPT for some SA tasks, specifically investigating its ability to handle polarity shifts, open-domain scenarios, and sentiment inference problems. In this work, we fine-tuned several small-scale large models released by the DeepSeek team using LoRA [30] technique, and investigated their effectiveness in sentiment classification of Chinese financial texts, further integrating them with time series data for stock prediction.

2.2 Deep models for financial time series forecasting

Financial time series forecasting has long been a highly sought-after field, with many researchers aiming to leverage deep learning techniques for better market predictions. Adv-ALSTM[31], a recurrent neural network enhanced by adversarial training is proposed to improve generalization in stock movement prediction via stochastic perturbations. A CNN-Transformer fusion model (CTTS)[32] is developed for intraday stock price change prediction, effectively capturing multi-scale temporal dependencies and outperforming statistical and deep learning baselines on SP 500 data. MAGNN [33], a multi-modality graph neural network, utilizing heterogeneous graphs and a two-phase attention mechanism to enhance interpretability and accuracy in financial time series prediction. TRNN [34] enhances time series data input, introduces an extrema compression algorithm, optimizes neural network architecture, and compares RNN and LSTM for improved stock market prediction accuracy and efficiency. DTML [35], a Data-axis Transformer with Multi-Level contexts, to model asymmetric and dynamic inter-stock correlations, achieving state-of-the-art accuracy in stock movement prediction across multiple markets. PatchTST[17], based on a similar idea to ViT [36],

partitions entire time series into several subsequences, treating each subsequence as a token for the Transformer encoder. This approach reduces the model's spatiotemporal complexity while achieving a performance breakthrough and achieves state of-the-art results in 2023.

2.3 Mamba-based models

As an emerging architecture in 2024, Mamba[23] has become a powerful alternative to Transformers, thanks to its efficient computation, low memory requirements, and strong long-sequence modeling capabilities. It holds great potential for applications in time series forecasting, long-text processing, and other related tasks. Pióro et al. [37] and Anthony et al. [38] replaced the Transformer architecture in the Mixture of Experts (MoE) with the Mamba architecture, achieving a complete override of Mamba's and Transformer-MoE's performance. S-Mamba [39], a Mamba-based time series forecasting model, leveraging bidirectional state space modeling to enhance efficiency and accuracy while reducing computational overhead. Aligned with similar objectives, our work aims to investigate short-term financial time series forecasting by leveraging the state space model Mamba and a mixture of experts framework.

3 Preliminaries

3.1 Problem Definition

The problem we address is as follows: Given a multivariate stock time series input $\mathbf{X} \in \mathbb{R}^{L \times C}$, where L represents the number of time steps and C denotes the number of features (e.g., closing price, trading volume, transaction amount, etc.). Meanwhile, a series of financial news and information for corresponding time periods are given.The objective is to predict the future T time steps of stock prices, represented as $\mathbf{Y} \in \mathbb{R}^{T}$. Formally, given a prediction function $F(\mathbf{X})$, we have $\mathbf{Y} = F(\mathbf{X})$. Our goal is to determine the optimal function that minimizes the discrepancy between the predicted and actual values. This study aims to develop an efficient forecasting model for precise stock price prediction.

3.2 Large Language Model

Large language models (LLMs) play a crucial role in sentiment analysis by leveraging their deep understanding of natural language to extract and interpret sentiment from financial texts [40], such as news articles, social media discussions, and analyst reports. Their ability to capture complex linguistic patterns, contextual nuances, and implicit sentiment allows for more accurate classification of market opinions into categories such as bullish, bearish, or neutral. Unlike traditional sentiment analysis methods that rely on handcrafted lexicons or shallow machine learning models, LLMs can generalize across different financial contexts and adapt to domain-specific terminologies without extensive manual feature engineering. Additionally, their capacity for few-shot and zero-shot learning enables sentiment classification with minimal labeled data, making them highly effective in dynamic financial environments where new expressions and sentiments constantly emerge. By integrating sentiment scores derived from LLMs into

stock price prediction models, it becomes possible to incorporate investor psychology and market expectations, enhancing the predictive power beyond numerical time series alone.

3.3 Selective State Spaces

State Space Models (SSMs) [21] provide a powerful framework for capturing periodic processes by utilizing latent states to model the temporal evolution of sequential data. These models describe system dynamics through first-order differential equations, ensuring a structured transformation from input sequences to output sequences. Formally, the continuous-time formulation of an SSM is expressed as:

$$\frac{d}{dt}h(t) = Ah(t) + Bx(t), \quad y(t) = Ch(t) \tag{1}$$

where $h(t)$ represents the latent state at time t, $x(t)$ is the input signal, and $y(t)$ is the observed output. The matrices A, B, and C define the state transition dynamics, input-to-state mapping, and state-to-output mapping, respectively.

For practical implementations, SSMs are typically discretized using a step size Δ, leading to the discrete-time formulation:

$$h_t = Ah_{t-1} + Bx_t, \quad y_t = Ch_t \tag{2}$$

where h_t represents the discrete-time latent state at step t. By leveraging structured state transitions, SSMs can efficiently model sequential dependencies.

To enhance the ability to capture long-range dependencies, the Structured State Space Model (S4) introduces the HiPPO [41] initialization method, which imposes structural constraints on the state matrix A. This structured approach significantly improves the capacity for handling long-term dependencies in sequential data.

Building upon S4, the Mamba model further refines sequence modeling by incorporating data-dependent selection mechanisms and hardware-optimized parallel algorithms, thereby enhancing both computational efficiency and predictive performance. Deitailed structure can be seen in **Figure 1**.

3.4 Mamba Block

The Mamba Block utilizes a structured state space model (SSM) with input-dependent parameter discretization to enhance sequential modeling capabilities. It begins with a linear projection operation, which expands the hidden dimension to obtain two separate representations, denoted as $\mathbf{x}$ and $\mathbf{z}$. The projected features then pass through a convolutional layer followed by a SiLU activation function, producing a transformed representation $\mathbf{x}'$. Subsequently, the structured state matrix A, along with the associated parameters B and C, is computed. The model further refines its parameters by determining the discretization factor Δ dynamically based on the input, which is then applied to adjust the state space parameters. The Selective SSM mechanism is employed to compute the final state representation $\mathbf{y}$, ensuring that the model effectively captures long-range dependencies. Finally, a SiLU [42] activation function, combined with residual connections, is applied to generate the output $\mathbf{Y}$. As seen in

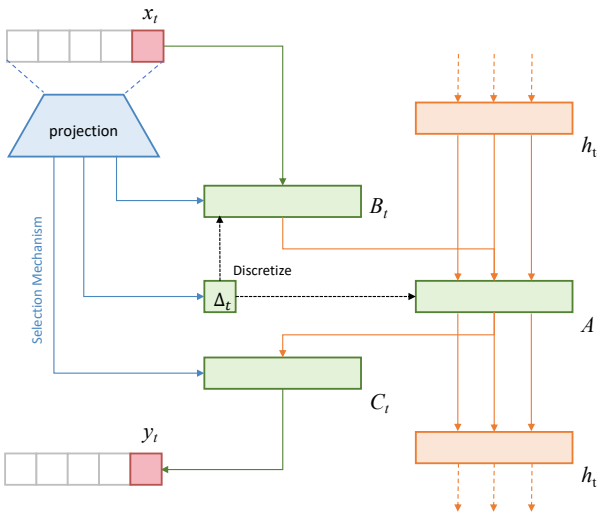


Fig. 1: The structure of selective SSM (Mamba).

Algorithm 1. By integrating these components, the Mamba architecture efficiently models sequential dependencies while maintaining computational efficiency superior to that of Transformer-based models.

Algorithm 1 The process of Mamba Block

Input: - $\mathbf{X} \in \mathbb{R}^{B \times V \times D}$: Input tensor, where B is the batch size, V is the sequence length,S is the dimension of state for mamba and D is the feature dimension. - Learnable parameters: $\mathbf{A}, \mathbf{B}, \mathbf{C}$ for the state-space model.
Output: - $\mathbf{Y} \in \mathbb{R}^{B \times V \times D}$: Output tensor with the same shape as input.

$\mathbf{x}, \mathbf{z} = \text{Linear}(\mathbf{X})$ ▷ $\mathbf{x}, \mathbf{z} \in \mathbb{R}^{B \times V \times ED}$
$\mathbf{x}' = \text{SiLU}(\text{Conv1D}(\mathbf{x}))$ ▷ $\mathbf{x}' \in \mathbb{R}^{B \times V \times ED}$
$\mathbf{B}, \mathbf{C} = \text{Linear}(\mathbf{x}'), \text{Linear}(\mathbf{x}')$ ▷ $\mathbf{B}, \mathbf{C} \in \mathbb{R}^{B \times V \times S}$
$\Delta = \text{Softplus}(\text{Parameter} + \text{Broadcast}(\text{Linear}(\mathbf{x}')))$ ▷ $\Delta \in \mathbb{R}^{B \times V \times D}$
$\mathbf{A}, \mathbf{B} = \text{Discretize}(\Delta, \mathbf{A}, \mathbf{B})$ ▷ $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{B \times V \times D \times S}$
$\mathbf{y} = \text{SelectiveSSM}(\mathbf{A}, \mathbf{B}, \mathbf{C})(\mathbf{x}')$ ▷ $\mathbf{y} \in \mathbb{R}^{B \times V \times ED}$
$\mathbf{y}' = \mathbf{y} \otimes \text{SiLU}(\mathbf{z})$ ▷ $\mathbf{y}' \in \mathbb{R}^{B \times V \times ED}$
$\mathbf{Y} = \text{Linear}(\mathbf{y}')$ ▷ $\mathbf{Y} \in \mathbb{R}^{B \times V \times D}$
return $\mathbf{Y}$

4 Methodology

We propose **Senti-MambaMoE**, a novel financial time series forecasting model that integrates Mamba and LLMs. We first perform sentiment classification on financial news titles using LLMs and compute the sentiment index for each corresponding date.

The sentiment index is then incorporated into the stock price dataset. For the input stock price sequence, we employ **MambaMoE** for modeling. The sequence is first normalized and then divided into patches, which are subsequently encoded. The encoding process involves intra-patch Mamba experts and inter-patch Mamba experts to capture both local and global dependencies. Finally, a linear layer decodes the representation to predict future stock prices, followed by inverse normalization to obtain the final output. The whole process our proposed method for stock prediction is illustrated in **Figure 2**.

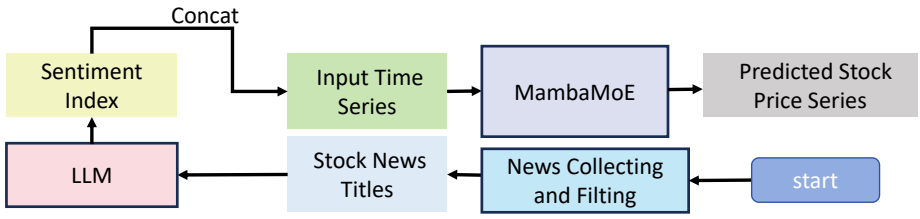


Fig. 2: Overview of Senti-MambaMoE.

4.1 Sentiment Analysis using Large Language Models

Large language models (LLMs) exhibit strong sentiment analysis capabilities due to their ability to capture contextual dependencies and nuanced linguistic patterns. High-Flyer Quant has recently introduced the DeepSeek R1 model and several distilled variants, incorporating architectural optimizations such as Multi-head Latent Attention (MLA) to enhance representation learning and Grouped-Query Attention (GQA) [43] to improve inference efficiency. For fine-tuning, DeepSeek models utilize Low-Rank Adaptation (LoRA) [30] to reduce trainable parameters and weight quantization techniques to optimize deployment on resource-constrained devices. Additionally, DeepSeek serves as a teacher model in knowledge distillation, transferring its knowledge to smaller student models while maintaining competitive performance.

In this study, we fine-tune multiple large language models using LoRA and quantization techniques to compare their effectiveness in Chinese sentiment classification. We employ structured input formats, converting raw text into .json files, and leverage `llamafactory` [44] for both fine-tuning and inference, ensuring efficient adaptation of the models to sentiment analysis tasks. The sentiment analysis process involves data preprocessing, where raw text is tokenized and normalized before being transformed into numerical embeddings. These embeddings are processed through transformer layers to extract contextual features, and the final hidden states are passed through a classification head to predict sentiment labels.

The financial news sentiment is classified into three categories: positive, negative, and neutral. For each trading day, we compute a sentiment index to quantify market sentiment based on the proportion of positive and negative sentiments. The index is

calculated as follows:

$$\text{Index} = \log\left(\frac{1 + \text{Positive}}{1 + \text{Negative}}\right) \tag{3}$$

where Positive and Negative represent the number of positive and negative sentiment classifications, respectively, on a given trading day. This logarithmic transformation ensures a stable and interpretable measure of market sentiment while mitigating the effects of extreme values.

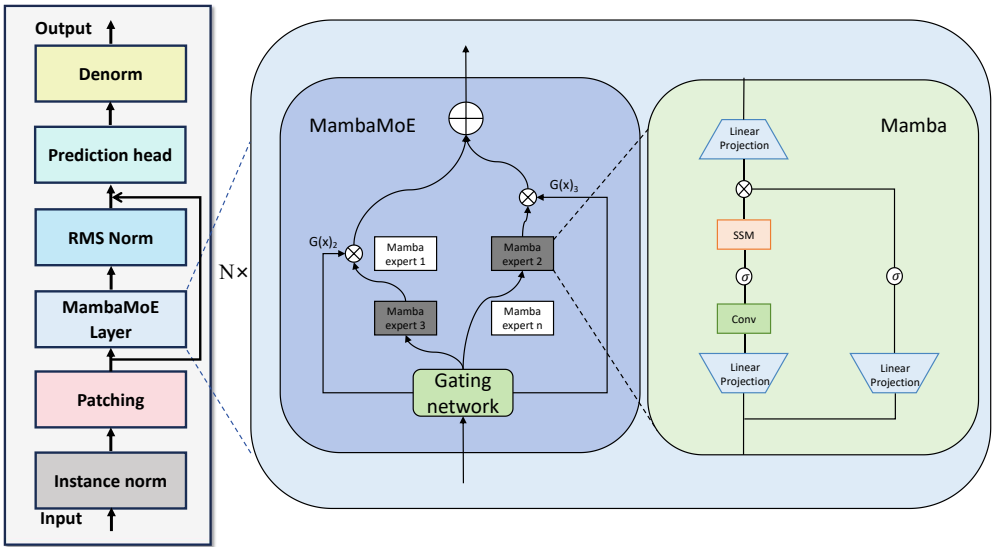


Fig. 3: The structure of MambaMoE. The right side of the figure details the components of MoE and Mamba.

4.2 MambaMoE for Time Series Forecasting

We propose **MambaMoE**, an advanced deep learning framework for short-term financial time series forecasting, as illustrated in **Figure 3**. MambaMoE seamlessly integrates the state-space model *Mamba* with a *Mixture of Experts (MoE)* mechanism, where multiple expert networks collaboratively learn temporal patterns. The number of experts is a hyperparameter. These experts are dynamically selected by a gating network, enabling adaptive modeling of financial market dynamics.

To effectively capture multi-scale dependencies within financial time series, we introduce a hierarchical expert architecture comprising two specialized expert groups:

- **Intra-Patch Experts**: These experts focus on short-term patterns, capturing localized trends, price fluctuations, and anomalies within individual patches.
- **Inter-Patch Experts**: These experts learn long-term dependencies by identifying broader trends, cyclic behaviors, and macroeconomic states (e.g., bull and bear markets).

By leveraging the MoE structure, MambaMoE dynamically selects the most relevant experts at each time step, improving prediction accuracy and generalization. The collaborative interaction between intra- and inter-patch experts ensures a balance between fine-grained feature extraction and global trend modeling, providing a robust forecasting framework.

4.2.1 Intra-Patch MoE

To model short-term dependencies, the **Intra-Patch MoE** mechanism operates at the local level. A gating network assigns weights to each intra-patch expert based on the input representation:

$$\mathbf{G}_1 = \text{softmax}(\mathbf{W}_g \mathbf{X}_{\text{patch}}) \in \mathbb{R}^{B \times N \times \text{pn} \times E} \tag{4}$$

where E represents the number of intra-patch experts. Each expert $\mathcal{E}_i$ processes the corresponding patch representation:

$$\mathbf{X}^i_{\text{patch}} = \mathcal{E}_i(\mathbf{X}_{\text{patch}}) \tag{5}$$

The final intra-patch representation is obtained through a weighted summation of expert outputs:

$$\overline{\mathbf{X}}_{\text{patch}} = \sum_{i=1}^{E} G_1^i \cdot \mathbf{X}^i_{\text{patch}} \tag{6}$$

4.2.2 Inter-Patch MoE

Once intra-patch features are extracted, they are treated as a new sequence for modeling long-term dependencies. The transformed sequence is represented as:

$$\mathbf{X}_{\text{inter}} = \text{Permute}(\overline{\mathbf{X}}_{\text{patch}}) \tag{7}$$

A second MoE gating network computes expert selection probabilities for inter-patch experts:

$$\mathbf{G}_2 = \text{softmax}(\mathbf{W}'_g \mathbf{X}_{\text{inter}}) \in \mathbb{R}^{B \times N \times D \times E'} \tag{8}$$

where E' denotes the number of inter-patch experts. Each inter-patch expert $\mathcal{E}_j$ learns cross-patch dependencies:

$$\mathbf{X}^j_{\text{inter}} = \mathcal{E}_j(\mathbf{X}_{\text{inter}}) \tag{9}$$

The final inter-patch representation is computed as:

$$\overline{\mathbf{X}}_{\text{inter}} = \sum_{j=1}^{E'} G_2^j \cdot \mathbf{X}^j_{\text{inter}} \tag{10}$$

After multiple stacked **MambaMoE** layers, a linear projection is applied for the final prediction:

$$\mathbf{Y} = \text{FlattenHead}(\overline{\mathbf{X}}_{\text{inter}}) \tag{11}$$

4.2.3 Expert Selection via Softmax Gating

In our model, we employ a softmax-based gating mechanism to dynamically assign weights to experts. Unlike Top-K selection, which chooses only a subset of experts, the softmax approach allows all experts to contribute to the final prediction, ensuring smooth transitions and stable training. This design choice is particularly suitable for financial time series forecasting, where market conditions change continuously rather than abruptly. Detailed comparison result of different expert selection strategies, including Top-K Gating and Softmax Gating, is provided in the ablation study.

The complete workflow of **MambaMoE** is summarized in **Algorithm 2**.

Algorithm 2 MambaMoE

Input: - $\mathbf{X} \in \mathbb{R}^{B \times L \times N}$: Input tensor, where B is the batch size, L is the sequence length, and N is the number of variables. - Learnable parameters: $\mathbf{A}, \mathbf{B}, \mathbf{C}$ for the state-space model. - pl: Patch length. - ps: Patch stride. - pn: Number of patches. - D: Hidden dimension of the model after patch embedding. - E: Number of experts in each Mixture-of-Experts (MoE) layer. - $\mathbf{G}_1, \mathbf{G}_2$: Gate score tensors used in the two MoE layers. - $\mathbf{H}_1^{(e)}, \mathbf{H}_2^{(e)}$: Expert outputs in the two MoE layers.
Output: - $\mathbf{Y} \in \mathbb{R}^{B \times T}$: Output tensor with the predicted values.

Step 1: Normalization and Patch Embedding
$\mathbf{X}_{\text{norm}} \leftarrow \text{Normalization}(\mathbf{X})$ ▷ Normalize input, $\mathbf{X}_{\text{norm}} \in \mathbb{R}^{B \times L \times N}$
$pl \leftarrow t$, $ps \leftarrow t/2$, $pn \leftarrow \left\lfloor \frac{L-pl}{ps} \right\rfloor + 2$
$\mathbf{X} \leftarrow \text{PatchEmbedding}(\mathbf{X}_{\text{norm}}, pl, ps, pn)$ ▷ Patch embedding, $\mathbf{X} \in \mathbb{R}^{B \times N \times pn \times D}$
Step 2: MoE Layers Processing
for $l \in$ Mamba Layers **do**
    Intra-Patch MoE (Expert Mixing within Patches)
    $\mathbf{G}_1 \leftarrow \text{Softmax}(\text{Linear}(\mathbf{X}))$ ▷ Compute gate scores, $\mathbf{G}_1 \in \mathbb{R}^{B \times N \times pn \times E}$
    $\mathbf{H}_1^{(e)} \leftarrow \text{Mamba}(\mathbf{X}) \quad \forall e \in \{1, \ldots, E\}$ ▷ Expert outputs
    $\mathbf{X} \leftarrow \sum_{e=1}^{E} \mathbf{G}_1^{(e)} \cdot \mathbf{H}_1^{(e)}$ ▷ Weighted sum, $\mathbf{X} \in \mathbb{R}^{B \times L \times pn \times D}$
    $\mathbf{X} \leftarrow \text{RMSNorm}(\mathbf{X}) + \mathbf{X}$ ▷ Apply residual connection
    Inter-Patch MoE (Expert Mixing across Patches)
    $\mathbf{X} \leftarrow \mathbf{X}.\text{permute}(0, 2, 1, 3)$ ▷ Rearrange shape to $\mathbb{R}^{B \times pn \times N \times D}$
    $\mathbf{G}_2 \leftarrow \text{Softmax}(\text{Linear}(\mathbf{X}))$ ▷ Compute gate scores, $\mathbf{G}_2 \in \mathbb{R}^{B \times pn \times N \times E}$
    $\mathbf{H}_2^{(e)} \leftarrow \text{Mamba}(\mathbf{X}) \quad \forall e \in \{1, \ldots, E\}$ ▷ Expert outputs
    $\mathbf{X} \leftarrow \sum_{e=1}^{E} \mathbf{G}_2^{(e)} \cdot \mathbf{H}_2^{(e)}$ ▷ Weighted sum, $\mathbf{X} \in \mathbb{R}^{B \times pn \times N \times D}$
    $\mathbf{X} \leftarrow \text{RMSNorm}(\mathbf{X}) + \mathbf{X}$ ▷ Apply residual connection
    $\mathbf{X} \leftarrow \mathbf{X}.\text{permute}(0, 2, 1, 3)$ ▷ Rearrange back to $\mathbb{R}^{B \times N \times pn \times D}$
end for
Step 3: Prediction Head
$\mathbf{Y} \leftarrow \text{FlattenHead}(\mathbf{X})$ ▷ Flatten for final prediction, $\mathbf{Y} \in \mathbb{R}^{B \times T}$
$\mathbf{Y} \leftarrow \text{DeNormalization}(\mathbf{X})$ ▷ DeNormalization for final prediction, $\mathbf{Y} \in \mathbb{R}^{B \times T}$
return $\mathbf{Y}$ ▷ Return the predicted output

5 Experiments

5.1 Datasets

All news headlines and articles were sourced from East Money. Using web scraping techniques, we collected article headlines published by official news accounts in the stock forums of Kweichow Moutai (600519.SH) from the liquor industry, CATL (300750.SZ) from the new energy vehicle sector, and Tencent Holdings (00700.HK) from the technology sector. The dataset spans from February 1, 2019, to March 1, 2025, and contains a total of 39,578 records, as presented in Table 1. We randomly selected approximately 10,000 samples from the Moutai news dataset and manually annotated them into three categories: positive, negative, and neutral. This labeled dataset was then used to fine-tune the large model.

Table 1: Text datasets description.

Stock code	amount	description
600519.SH	12394	liquor industry
300750.SZ	12025	new energy vehicle
00700.HK	15159	technology&Internet

For stock price data, we used the Tushare API to extract historical price time series for all trading days within the same period. This dataset includes key financial indicators such as opening price, highest price, closing price, trading volume, and turnover, as outlined in **Table 2**.

Table 2: Stock datasets description.

Stock code	time steps	start date	end date	frequency
600519.SH	1372	2019-06-06	2025-01-27	daily
300750.SZ	1475	2019-02-01	2025-03-07	daily
00700.HK	1493	2019-02-01	2025-02-28	daily

5.2 Baselines

We fine-tuned seven models on the manually annotated part of the Kweichow Moutai news dataset. All models were sourced from the ModelScope. The models that were fine-tuned include: Qwen2.5-1.5b, DeepSeek-R1-distill-Qwen-1.5b, Meta-llama-3-8b, DeepSeek-R1-Distill-Llama3-8B, DeepSeek-LLM-7b-base, FinBERT, and BERT.

For the stock sequence prediction task, in addition to our proposed Senti-MambaMoE model, we also trained PatchTST [17], vanilla Mamba [23] , TSMixer [45]and BiLSTM as baseline models for comparison.

5.3 Experimental setup

For fine-tuning the pre-trained large models, we applied 4-bit quantization with a LoRA rank of 8. The maximum input token length was set to 128, with a learning rate of 0.0005, cosine learning rate decay, and a temperature coefficient of 0.9. We trained for 5 epochs with a batch size of 12. The dataset was split into training, validation, and test sets in an 8:1:1 ratio. To address the issue of missing sentiment indices on certain trading days, we employ an exponential smoothing approach. Specifically, for a given missing sentiment index S_t, we estimate its value using the weighted sum of the most recent five available sentiment indices.

$$S_t = \frac{\sum_{i=1}^{5} w_i S_{t-i}}{\sum_{i=1}^{5} w_i}, \quad w_i = \alpha(1-\alpha)^{i-1} \tag{12}$$

where S_{t-i} represents the sentiment index of the i-th most recent trading day with available data, and w_i is the corresponding weight. The smoothing factor α controls the influence of past observations, ensuring a balanced interpolation that captures sentiment trends effectively. α is set to 0.9 in our experiments.

For the stock prediction task, We employ a sliding window approach to segment the input data, using a 48-step window to forecast stock prices for the next 12 steps. To address the non-stationarity of stock data, we normalize the input sequences before modeling and apply inverse normalization to the outputs before generating the final predictions. The dataset was split into training, validation, and test sets in an 8:1:1 ratio. For all models, hyperparameter search was used for the hidden layer dimensions 8, 16, 32, 64 and learning rates 0.0001, 0.00005, 0.00005. The Mamba model was configured with 3 layers, 2 experts, a hidden state size of 8, and a convolution kernel size of 4. We set the batch size to 128, and training was conducted for 20 epochs with an early stopping tolerance of 3. The patch length was set to 12, with an input length of 48 and a prediction length of 12. Further details on different parameter settings can be found in the parameter sensitivity section. All the experiments are performed on an i9-13900HX and a RTX-4080 GPU. The best metrics are highlighted in bold During the training process, we used the MSE loss function, and additionally, selected R-squared and MAPE as evaluation metrics.

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \tag{13}$$

$$\text{MAPE} = \frac{1}{n}\sum_{i=1}^{n}\left|\frac{y_i - \hat{y}_i}{y_i}\right| \times 100 \tag{14}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \tag{15}$$

5.4 Result of sentiment analysis

Table 3 presents the performance of the fine-tuned large model on sentiment analysis, demonstrating that **DeepSeek-R1-Distill-Llama3-8B** achieves optimal results

in Chinese text classification tasks. Notably, the distilled model consistently outperforms its non-distilled counterpart, highlighting the benefits of knowledge distillation in enhancing classification accuracy while reducing computational costs. Furthermore, we observe a clear correlation between model performance and parameter size, where larger models tend to exhibit stronger classification capabilities.To show detailed classification effectiveness, we present the confusion matrix of sentiment analysis in **Figure 4**.

Given these findings, we employ the fine-tuned **DeepSeek-R1-Distill-Llama3-8B** for sentiment analysis in subsequent stock prediction tasks. Specifically, we utilize the model to compute sentiment indices for the three selected stocks, leveraging their respective sentiment trends to explore potential links between market sentiment and stock price movements. To provide a more intuitive understanding, we calculate the daily sentiment index for each stock and visualize the sentiment index of 600519.SH alongside its stock performance in **Figure 5**. This comparison provides valuable insights into the correlation between investor sentiment and stock market trends.

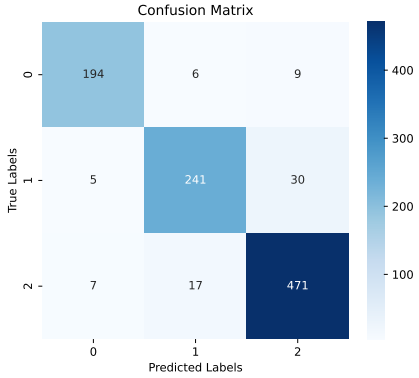


Fig. 4: Confusion matrix of sentiment analysis.

Table 3: Result of LLMs sentiment analysis.

Model	accuracy
Qwen2.5-1.5b	0.898
DeepSeek-R1-distill-Qwen-1.5b	0.912
Meta-llama-3-8b	0.911
DeepSeek-R1-Distill-Llama3-8B	**0.924**
DeepSeek-LLM-7b-base	0.920
FinBERT	0.893
BERT	0.856

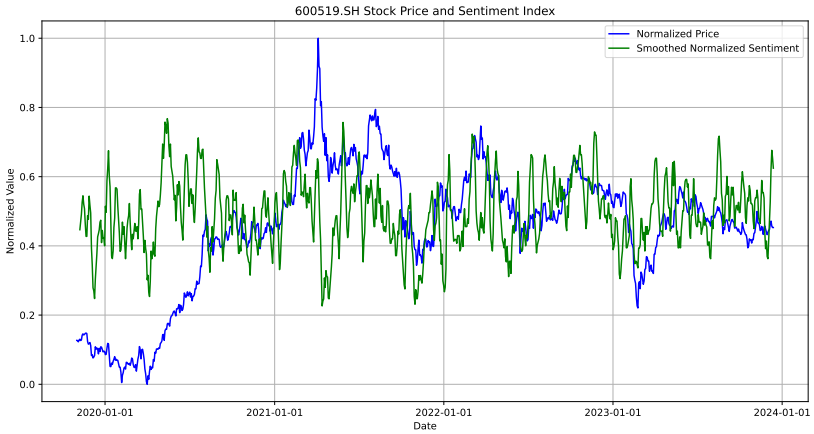


Fig. 5: Visualization of smoothed sentiment index and normalized stock prices of 600519.SH.

5.5 Result of prediction

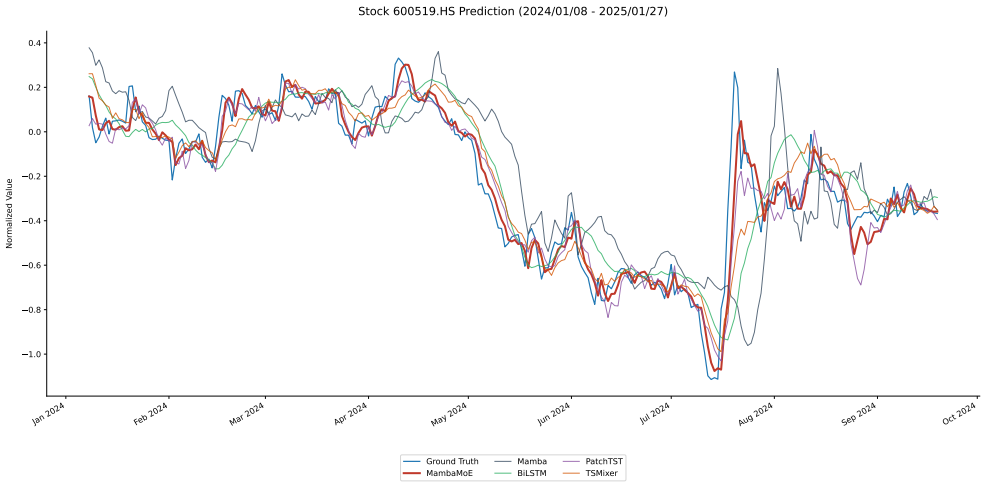


Fig. 6: Visualization of stock prediction result of 600519.SH.

Table 4 demonstrates that, across the three stock datasets, our model outperforms all other baseline models, with the best results highlighted in bold. **Figure 6** presents the prediction results of all baseline models on 600519.SH. In our comparative experiments, we selected the original Mamba, to highlight the superiority of our model's improvements. We also chose the state-of-the-art Transformer-based time series model

Table 4: Performance of different models on Stock datasets. The lookback window is set to 48 and the prediction length is set to 12.

	Ours			Mamba			PatchTST			TSMixer			BiLSTM		
Dataset	MSE	MAPE	R^2	MSE	MAPE	R^2	MSE	MAPE	R^2	MSE	MAPE	R^2	MSE	MAPE	R^2
600519.SH	0.0402	3.4980	0.6251	0.0715	4.688	0.4024	0.0436	4.9312	0.5999	0.0425	4.6744	0.6231	0.0528	4.8483	0.5196
600519.SH + SI	**0.0398**	**3.3777**	**0.6314**	0.0704	4.5291	0.4174	0.0424	4.8822	0.6072	0.0414	4.5260	0.6168	0.0521	4.8432	0.5179
300750.SZ	0.0058	0.1263	0.8710	0.0106	0.1769	0.7654	0.0063	0.1310	0.8658	0.0061	0.1226	0.8657	0.0077	0.1432	0.8308
300750.SZ + SI	**0.0057**	**0.1230**	**0.8747**	0.0090	0.1601	0.8008	0.0060	0.1275	0.8663	0.0061	0.1266	0.8657	0.0071	0.1348	0.8419
00700.HK	0.0323	0.9227	0.8528	0.0692	1.6797	0.6849	0.0340	0.9211	0.8448	0.0351	0.9736	0.8398	0.0436	1.024	0.8011
00700.HK + SI	**0.0318**	**0.8802**	**0.8551**	0.0565	1.3769	0.7427	0.0339	0.9363	0.8456	0.0338	0.7567	0.8458	0.0422	1.0236	0.8078

PatchTST, the MLP-based model TSMixer, and the recurrent neural network model BiLSTM for comparison. Overall, predictions incorporating sentiment data generally outperform those without it, suggesting that incorporating news sentiment can enhance stock price forecasting. Our model achieves significant improvements over the original Mamba by 44.3% (on MSE), outperforming Transformer-based models by 6.3% and LSTM-based models by 23.7%, demonstrating its effectiveness in capturing both short-term fluctuations and long-term dependencies. By employing two expert groups—one for extracting local information within patches and the other for capturing global dependencies between patches—our model not only consumes fewer resources compared to Transformer-based models but also enhances Mamba's capability in long-range sequence modeling. These results highlight the advantages of our approach in financial time series forecasting.

5.6 Ablation study

Mixture of Experts (MoE) enhances prediction capabilities by introducing multiple expert networks and dynamically adjusting their contributions through a gating mechanism. To illustrate MoE's role in time series forecasting, we constructed a univariate time series characterized by a long-term upward trend, multiple nested periodicities, Gaussian noise, and three abrupt change points. A MoE framework, combined with simple linear layers, was employed for prediction. As shown in **Figure 7**, MoE effec-

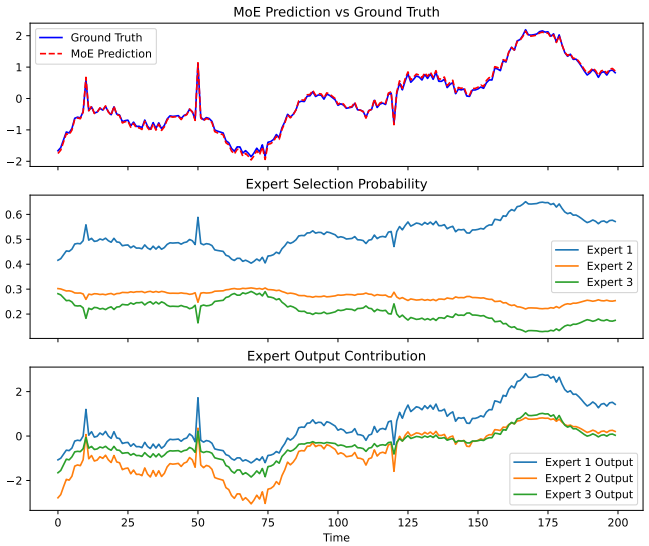


Fig. 7: Demonstration of how MoE works on time series forecasting.

tively models complex time series, demonstrating strong performance in prediction

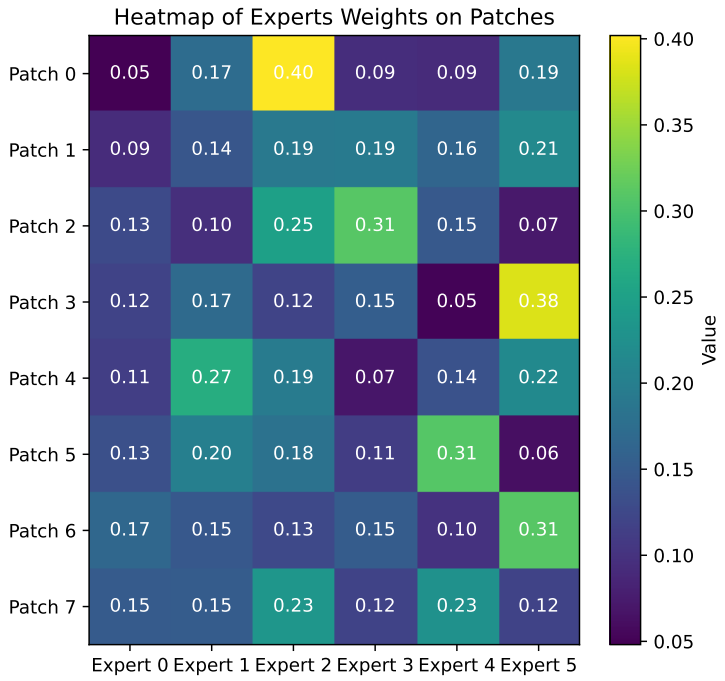


Fig. 8: Visualization of Experts weights on patches. This is a snapshot obtained during the first epoch of training on the 600519.SH dataset.

accuracy, dynamic expert selection, and expert contribution heterogeneity. The predicted curve closely aligns with the actual sequence, capturing trends, periodicity, and local fluctuations with high fidelity. Even at abrupt change points, MoE adapts well, reflecting its ability to handle diverse data patterns.

The gating network dynamically assigns different weights to experts across time steps, ensuring the activation of the most suitable expert based on local data characteristics. During stable periods, a dominant expert is consistently selected, whereas around change points, multiple experts contribute significantly. This adaptive selection mechanism allows MoE to refine its prediction strategy in response to shifting patterns. In stock prediction for 600519.SH, we visualize the expert weight distribution of Inter-Patch MoE during training in **Figure 8**.

In this section, we also evaluate the impact of different components on the performance of the model using the 600519.SH dataset, as shown in Table 5. The results indicate that the full **senti-MambaMoE** model performs the best, achieving the lowest MSE (0.0398), MAPE (3.3772), and the highest R^2 (0.6314), highlighting the effectiveness of incorporating sentiment analysis and the Mixture of Experts (MoE) approach.

When sentiment analysis is removed (**w/o sentiment**), the model shows a slight increase in MSE and MAPE and a small decrease in R^2, suggesting that sentiment information contributes to improved performance and the same phenomenon can be seen in Table 4 too. Similarly, the removal of MoE (**w/o MoE**) results in a noticeable decline in performance, with higher MSE and MAPE and a lower R^2, underscoring the importance of the MoE component in capturing complex patterns. The variants **w/o intra-patch MoE** show minor changes compared to the full model, indicating that while the intra-patch MoE has a smaller effect, it still contributes to overall performance. Additionally, when the entire Mamba architecture is removed (**w/o Mamba**) and replaced with a standard MLP-based backbone, the performance significantly drops, with an MSE increase to 0.0448 and a lower R^2 (0.5859). This highlights the advantages of structured state space modeling in financial time series forecasting. Furthermore, we experiment with a different gating strategy by restricting the MoE selection to only the top-2 highest-scoring experts (**top-2 experts gating**). This setting results in an MSE of 0.0418 and an even larger increase in MAPE (4.1697), suggesting that allowing a more flexible expert selection (as in the full model) leads to better performance. Finally, the **vanilla Mamba** model, which lacks both patching and MoE, performs the worst, with significantly higher error metrics (MSE = 0.0704, MAPE = 4.5291) and the lowest R^2 (0.4174).

Table 5: Ablations on 600519.SH dataset.

Model	MSE	MAPE	$\mathbf{R^2}$
senti-MambaMoE	**0.0398**	**3.3772**	**0.6314**
w/o sentiment	0.0402	3.4980	0.6251
w/o MoE	0.0410	3.8353	0.6129
w/o Mamba	0.0448	3.9889	0.5859
w/o intra-patch MOE	0.0400	3.4614	0.6261
w/o inter-patch MOE	0.0402	3.4657	0.6286
top-2 experts gating	0.0418	4.1697	0.6136
vanilla Mamba	0.0704	4.5291	0.4174

5.7 Parameter sensitivity

In this section, we investigate the impact of several hyperparameters in Senti-MambaMoE on model performance. Specifically, we examine the number of experts {1,2,3,4,5,6} and the input sequence length {24,36,48,60,72,96}.

As illustrated in Table 6, the optimal number of experts varies across different stock datasets, which may be attributed to the distinct internal patterns within each stock. Additionally, the input length plays a crucial role in shaping the model's predictive performance. For time series with strong periodicity, increasing the input length often improves predictive performance. However, we did not observe this effect in stock price forecasting, likely due to the inherently high noise and lack of clear periodic patterns in stock movements. To analyze the impact of input length, we evaluate all baseline models across three datasets, as shown in **Table 7**.

Table 6: Performance of different numbers of experts on stock datasets.

	600519.SH			300750.SZ			00700.HK		
experts	MSE	MAPE	R^2	MSE	MAPE	R^2	MSE	MAPE	R^2
1	0.0401	3.5234	0.6292	0.0442	1.1978	0.8121	0.0059	0.1393	0.8808
2	0.0401	3.4372	0.6291	0.0452	1.4000	0.8093	0.0057	0.1320	0.8869
3	0.0413	3.6089	0.6180	0.0452	1.3913	0.8096	**0.0055**	**0.1296**	**0.8904**
4	**0.0398**	3.3772	**0.6314**	0.0434	1.2474	0.8172	0.0058	0.1348	0.8846
5	0.0442	4.4149	0.5916	**0.0405**	**1.1967**	**0.8293**	0.0059	0.1384	0.8826
6	0.0400	**3.1772**	0.6301	0.0454	1.4273	0.8085	0.0059	0.1371	0.8833

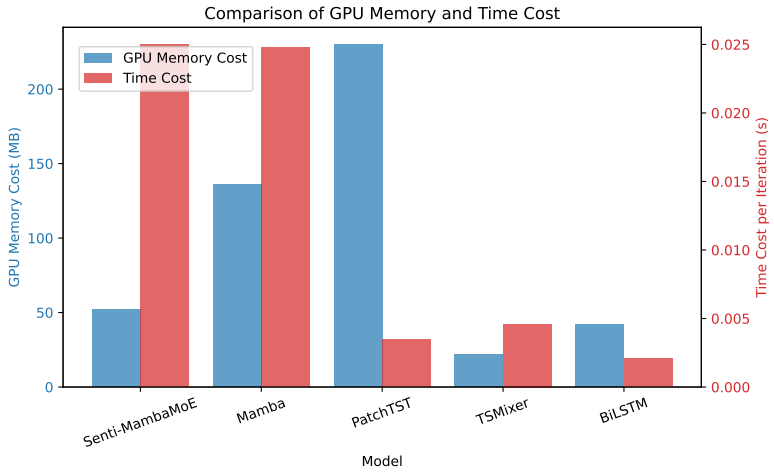


Fig. 9: Comparison of GPU memory and time cost. The lookback window is set to 48 and prediction length is set to 12.

5.8 Computation Efficiency

In this section, we explore the efficiency and memory consumption of senti-MambaMoE in Figure 9 . One of the key advantages of models based on the Mamba architecture is their O(n) memory complexity. Although memory usage and training speed have minimal impact on stock price prediction, we conducted a comparative analysis for reference, considering the model's potential applicability to other time series forecasting tasks. Overall, the Mamba model has lower memory consumption but higher computational latency. This is because it processes sequences recursively, avoiding the high memory overhead of global attention. However, since each time step requires matrix operations and state updates, the computation is executed sequentially, resulting in longer processing times.

Table 7: Varying input lengths impact on 600519.SH. The lookback window is set to 24,36,48,60,72,96 and prediction length is set to 12.

	Ours			Mamba			PatchTST			TSMixer			BiLSTM		
Input Length	**MSE**	**MAPE**	$\mathbf{R^2}$	**MSE**	**MAPE**	$\mathbf{R^2}$	**MSE**	**MAPE**	$\mathbf{R^2}$	**MSE**	**MAPE**	$\mathbf{R^2}$	**MSE**	**MAPE**	$\mathbf{R^2}$
24	**0.0403**	**2.786**	**0.6262**	0.0525	4.5367	0.5144	0.0405	2.8379	0.6248	0.0410	3.8018	0.6202	0.0474	4.4887	0.5611
36	**0.0413**	**3.2076**	**0.6201**	0.0624	4.2365	0.4226	0.0429	3.7346	0.6029	0.0446	4.1998	0.5875	0.0545	4.9067	0.4960
48	**0.0398**	**3.3777**	**0.6314**	0.0704	4.5291	0.4174	0.0424	4.882	0.6072	0.0414	4.5260	0.6168	0.0521	4.8432	0.5179
60	**0.0424**	**3.5437**	**0.6081**	0.0688	4.5054	0.3632	0.0460	3.7048	0.5746	0.0494	4.5626	0.5426	0.0521	5.2102	0.5178
72	0.0416	**3.4552**	**0.6199**	0.0803	4.6130	0.2571	**0.0415**	3.5267	0.6158	0.0434	0.1415	0.5983	0.0576	5.0463	0.4674
96	0.0411	3.7339	0.6195	0.1069	3.8810	0.2444	**0.0407**	**3.3711**	**0.6235**	0.0490	4.2893	0.5469	0.0561	4.6164	0.4807

6 Conclusion and future directions

In this study, we propose a novel two-stage financial time series forecasting model **Senti-MambaMoE** that integrates large language model (LLM)-based sentiment analysis with a Mamba-based mixture-of-experts (MoE) architecture. By embedding sentiment signals into time series data and leveraging Mamba's state space modeling capabilities, our approach effectively captures both short-term fluctuations and long-term dependencies. To enhance predictive accuracy, we fine-tune advanced quantized LLMs for Chinese financial sentiment analysis, develop a state space model that seamlessly incorporates sentiment data into time series forecasting, and introduce a hierarchical MoE architecture that improves feature extraction at both local and global scales. Extensive experiments on Chinese A-share and Hong Kong stock market datasets demonstrate the superiority of our model, achieving performance gains of 6.3% over Transformer-based models, 23.7% over BiLSTM, and 43.4% over the original Mamba architecture. These results highlight the potential of mamba-based models for real-world financial forecasting applications.

While our model demonstrates promising results, several areas warrant further exploration. Our current sentiment embeddings rely on news headlines from the Eastmoney stock forum, and future research could investigate integrating multiple sentiment sources, such as real-time social media discussions, analyst reports, and broader market indicators, to capture a more comprehensive market sentiment. Additionally, future work will explore combining manually set and dynamically selected experts, leveraging reinforcement learning-based gating mechanisms to enhance adaptability and robustness across different stock sectors.

Authorship contribution

Renjie Wang: Conceptualization, Methodology, Software, Investigation, Formal analysis, Writing - Original Draft. **Minghui Sun:** Project administration, Supervision. **Limin Wang:** Resources.

Competing interest declaration

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

All datasets used in this study are publicly available.

Funding declaration

This study has been partially supported by Program of Science and Technology Development Plan of Jilin Province of China under Grant 20220201147GX, and Grant 20240101374JC, and Shenzhen Science and Technology Program (JCYJ20230807150300001).

Ethics approval

Not applicable.

References

[1] Mondal, P., Shit, L., Goswami, S.: Study of effectiveness of time series modeling (arima) in forecasting stock prices. International Journal of Computer Science, Engineering and Applications **4**(2), 13 (2014) https://doi.org/10.5121/ijcsea.2014.4202

[2] Lu, W., Li, J., Li, Y., Sun, A., Wang, J.: A cnn-lstm-based model to forecast stock prices. Complexity **2020**(1), 6622927 (2020) https://doi.org/10.1155/2020/6622927

[3] Md, A.Q., Kapoor, S., A.V., C.J., Sivaraman, A.K., Tee, K.F., H., S., N., J.: Novel optimization approach for stock price forecasting using multi-layered sequential lstm. Applied Soft Computing **134**, 109830 (2023) https://doi.org/10.1016/j.asoc.2022.109830

[4] Hochreiter, S., Schmidhuber, J.: Long short-term memory. Neural computation **9**(8), 1735–1780 (1997) https://doi.org/10.1162/neco.1997.9.8.1735

[5] Vo, Q.-H., Nguyen, H.-T., Le, B., Nguyen, M.-L.: Multi-channel lstm-cnn model for vietnamese sentiment analysis. In: 2017 9th International Conference on Knowledge and Systems Engineering (KSE), pp. 24–29 (2017). https://doi.org/10.1109/KSE.2017.8119429 . IEEE

[6] Kim, Y.: Convolutional Neural Networks for Sentence Classification (2014). https://arxiv.org/abs/1408.5882

[7] Devlin, J., Chang, M.-W., Lee, K., Toutanova, K.: Bert: Pre-training of deep bidirectional transformers for language understanding. In: Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (long and Short Papers), pp. 4171–4186 (2019). https://doi.org/10.18653/v1/N19-1423

[8] Liu, Z., Huang, D., Huang, K., Li, Z., Zhao, J.: Finbert: A pre-trained financial language representation model for financial text mining. In: Proceedings of the Twenty-ninth International Conference on International Joint Conferences on Artificial Intelligence, pp. 4513–4519 (2021). https://www.ijcai.org/proceedings/2020/0622.pdf

[9] Huang, A.H., Wang, H., Yang, Y.: Finbert: A large language model for extracting information from financial text. Contemporary Accounting Research **40**(2), 806–841 (2023) https://doi.org/10.1111/1911-3846.12832

[10] Ouyang, L., Wu, J., Jiang, X., Almeida, D., Wainwright, C., Mishkin, P., Zhang, C., Agarwal, S., Slama, K., Ray, A., *et al.*: Training language models to follow instructions with human feedback. Advances in neural information processing systems **35**, 27730–27744 (2022)

[11] DeepSeek-AI, Guo, D., Yang, D., Zhang, H., et al.: DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning (2025). https://arxiv.org/abs/2501.12948

[12] Dai, D., Deng, C., Zhao, C., Xu, R.X., Gao, H., Chen, D., Li, J., Zeng, W., Yu, X., Wu, Y., Xie, Z., Li, Y.K., Huang, P., Luo, F., Ruan, C., Sui, Z., Liang, W.: DeepSeekMoE: Towards Ultimate Expert Specialization in Mixture-of-Experts Language Models (2024). https://arxiv.org/abs/2401.06066

[13] Cho, K., Merrienboer, B., Gulcehre, C., Bahdanau, D., Bougares, F., Schwenk, H., Bengio, Y.: Learning Phrase Representations using RNN Encoder-Decoder for Statistical Machine Translation (2014). https://arxiv.org/abs/1406.1078

[14] Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, L.u., Polosukhin, I.: Attention is all you need. In: Guyon, I., Luxburg, U.V., Bengio, S., Wallach, H., Fergus, R., Vishwanathan, S., Garnett, R. (eds.) Advances in Neural Information Processing Systems, vol. 30. Curran Associates, Inc., ??? (2017). https://proceedings.neurips.cc/paper_files/paper/2017/file/3f5ee243547dee91fbd053c1c4a845aa-Paper.pdf

[15] Zhang, Y., Yan, J.: Crossformer: Transformer utilizing cross-dimension dependency for multivariate time series forecasting. In: The Eleventh International Conference on Learning Representations (2023). https://openreview.net/forum?id=vSVLM2j9eie

[16] Zhou, T., Ma, Z., Wen, Q., Wang, X., Sun, L., Jin, R.: Fedformer: Frequency enhanced decomposed transformer for long-term series forecasting. In: International Conference on Machine Learning, pp. 27268–27286 (2022). PMLR. https://proceedings.mlr.press/v162/zhou22g.html

[17] Nie, Y., Nguyen, N.H., Sinthong, P., Kalagnanam, J.: A time series is worth 64 words: Long-term forecasting with transformers. arXiv preprint arXiv:2211.14730 (2022) https://doi.org/10.48550/arXiv.2211.14730

[18] Zeng, A., Chen, M., Zhang, L., Xu, Q.: Are transformers effective for time series forecasting? In: Proceedings of the AAAI Conference on Artificial Intelligence, vol. 37, pp. 11121–11128 (2023). https://doi.org/10.1609/aaai.v37i9.26317

[19] Wang, S., Wu, H., Shi, X., Hu, T., Luo, H., Ma, L., Zhang, J.Y., Zhou, J.: Timemixer: Decomposable multiscale mixing for time series forecasting. arXiv preprint arXiv:2405.14616 (2024) https://doi.org/10.48550/arXiv.2405.14616

[20] Ekambaram, V., Jati, A., Nguyen, N., Sinthong, P., Kalagnanam, J.: Tsmixer: Lightweight mlp-mixer model for multivariate time series forecasting. In: Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining, pp. 459–469 (2023). https://doi.org/10.48550/arXiv.2306.09364

[21] Gu, A., Goel, K., Ré, C.: Efficiently Modeling Long Sequences with Structured State Spaces (2022). https://arxiv.org/abs/2111.00396

[22] Smith, J.T.H., Warrington, A., Linderman, S.W.: Simplified State Space Layers for Sequence Modeling (2023). https://arxiv.org/abs/2208.04933

[23] Gu, A., Dao, T.: Mamba: Linear-time sequence modeling with selective state spaces. arXiv preprint arXiv:2312.00752 (2023) https://doi.org/10.48550/arXiv.2312.00752

[24] Wang, J., Yu, L.-C., Lai, K.R., Zhang, X.: Dimensional sentiment analysis using a regional cnn-lstm model. In: Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (volume 2: Short Papers), pp. 225–230 (2016). https://aclanthology.org/P16-2037.pdf

[25] Yang, J., Yu, Y., Niu, D., Guo, W., Xu, Y.: ConFEDE: Contrastive feature decomposition for multimodal sentiment analysis. In: Rogers, A., Boyd-Graber, J., Okazaki, N. (eds.) Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), pp. 7617–7630. Association for Computational Linguistics, Toronto, Canada (2023). https://doi.org/10.18653/v1/2023.acl-long.421 . https://aclanthology.org/2023.acl-long.421/

[26] Lu, X., Qiu, J., Yang, Y., Zhang, C., Lin, J., An, S.: Large language model-based bidding behavior agent and market sentiment agent-assisted electricity price prediction. IEEE Transactions on Energy Markets, Policy and Regulation, 1–13 (2024) https://doi.org/10.1109/TEMPR.2024.3518624

[27] Wang, X., Feng, M., Qiu, J., Gu, J., Zhao, J.: From news to forecast: Integrating event analysis in llm-based time series forecasting with reflection. In: Globerson, A., Mackey, L., Belgrave, D., Fan, A., Paquet, U., Tomczak, J., Zhang, C. (eds.) Advances in Neural Information Processing Systems, vol. 37, pp. 58118–58153. Curran Associates, Inc., ??? (2024). https://proceedings.neurips.cc/paper_files/paper/2024/file/6aef8bffb372096ee73d98da30119f89-Paper-Conference.pdf

[28] Chen, Q., Kawashima, H.: Stock price prediction using llm-based sentiment analysis. In: 2024 IEEE International Conference on Big Data (BigData), pp. 4846–4853 (2024). https://doi.org/10.1109/BigData62323.2024.10825946

[29] Wang, Z., Xie, Q., Feng, Y., Ding, Z., Yang, Z., Xia, R.: Is chatgpt a good sentiment analyzer? a preliminary study. arXiv preprint arXiv:2304.04339 (2023) https://doi.org/10.48550/arXiv.2304.04339

[30] Hu, E.J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Wang, L., Chen, W., *et al.*: Lora: Low-rank adaptation of large language models. ICLR **1**(2), 3 (2022)

[31] Feng, F., Chen, H., He, X., Ding, J., Sun, M., Chua, T.-S.: Enhancing Stock Movement Prediction with Adversarial Training (2019). https://arxiv.org/abs/1810.09936

[32] Zeng, Z., Kaur, R., Siddagangappa, S., Rahimi, S., Balch, T., Veloso, M.: Financial time series forecasting using cnn and transformer. arXiv preprint arXiv:2304.04912 (2023)

[33] Cheng, D., Yang, F., Xiang, S., Liu, J.: Financial time series forecasting with multi-modality graph neural network. Pattern Recognition **121**, 108218 (2022) https://doi.org/10.1016/j.patcog.2021.108218

[34] Lu, M., Xu, X.: Trnn: An efficient time-series recurrent neural network for stock price prediction. Information Sciences **657**, 119951 (2024) https://doi.org/10.1016/j.ins.2023.119951

[35] Yoo, J., Soun, Y., Park, Y.-c., Kang, U.: Accurate multivariate stock movement prediction via data-axis transformer with multi-level contexts. In: Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery & Data Mining. KDD '21, pp. 2037–2045. Association for Computing Machinery, New York, NY, USA (2021). https://doi.org/10.1145/3447548.3467297 . https://doi.org/10.1145/3447548.3467297

[36] Dosovitskiy, A.: An image is worth 16x16 words: Transformers for image recognition at scale. arXiv preprint arXiv:2010.11929 (2020) https://doi.org/10.48550/arXiv.2010.11929

[37] Pióro, M., Ciebiera, K., Król, K., Ludziejewski, J., Krutul, M., Krajewski, J., Antoniak, S., Miłoś, P., Cygan, M., Jaszczur, S.: MoE-Mamba: Efficient Selective State Space Models with Mixture of Experts (2024). https://arxiv.org/abs/2401.04081

[38] Anthony, Q., Tokpanov, Y., Glorioso, P., Millidge, B.: Blackmamba: Mixture of experts for state-space models. ArXiv **abs/2402.01771** (2024)

[39] Wang, Z., Kong, F., Feng, S., Wang, M., Yang, X., Zhao, H., Wang, D., Zhang, Y.: Is mamba effective for time series forecasting? Neurocomputing **619**, 129178 (2025) https://doi.org/10.1016/j.neucom.2024.129178

[40] Inserte, P.R., Nakhlé, M., Qader, R., Caillaut, G., Liu, J.: Large language model adaptation for financial sentiment analysis. arXiv preprint arXiv:2401.14777 (2024) https://doi.org/10.18653/v1/2023.finnlp-2.1"

[41] Gu, A., Dao, T., Ermon, S., Rudra, A., Ré, C.: Hippo: Recurrent memory with optimal polynomial projections. In: Larochelle, H., Ranzato, M., Hadsell, R., Balcan, M.F., Lin, H. (eds.) Advances in Neural Information Processing Systems, vol. 33, pp. 1474–1487. Curran Associates, Inc., ??? (2020). https://proceedings.neurips.cc/paper_files/paper/2020/file/102f0bb6efb3a6128a3c750dd16729be-Paper.pdf

[42] Elfwing, S., Uchibe, E., Doya, K.: Sigmoid-Weighted Linear Units for Neural Network Function Approximation in Reinforcement Learning (2017). https://arxiv.org/abs/1702.03118

[43] Ainslie, J., Lee-Thorp, J., De Jong, M., Zemlyanskiy, Y., Lebrón, F., Sanghai, S.: Gqa: Training generalized multi-query transformer models from multi-head checkpoints. arXiv preprint arXiv:2305.13245 (2023) https://doi.org/10.18653/v1/2023.emnlp-main.298

[44] Zheng, Y., Zhang, R., Zhang, J., Ye, Y., Luo, Z., Feng, Z., Ma, Y.: Llamafactory: Unified efficient fine-tuning of 100+ language models. arXiv preprint arXiv:2403.13372 (2024) https://doi.org/10.18653/v1/2024.acl-demos.38

[45] Chen, S.-A., Li, C.-L., Yoder, N., Arik, S.O., Pfister, T.: Tsmixer: An all-mlp architecture for time series forecasting. arXiv preprint arXiv:2303.06053 (2023) https://doi.org/10.48550/arXiv.2303.06053